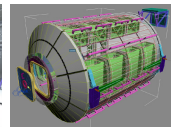




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Crew Maintenance Lessons Learned from ISS and Considerations for Future Manned Missions

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Nomenclature

BBA	Base Ballast Assembly
EVA	Extra-vehicular Activity
GLA	General Luminaire Assembly
ISS	International Space Station
IVA	Intra-vehicular Activity
I-level	Intermediate-level
LHA	Lamp Housing Assembly
MDM	Multiplexer/De-Multiplexer
MPLM	Multi-Purpose Logistics Module
MTBF	Mean Time Between Failures
ORU	Orbital Replacement Unit
RS	Russian Segment
USOS	United States On-orbit Segment

1 Abstract

The future of manned spaceflight is expected to return humans to the moon and eventually beyond to Mars. This will present unprecedented challenges for spaceflight operations, mostly due to the significantly greater distances and durations required for such missions. One of these challenges is crew maintenance of the spacecraft. Design studies have shown most likely durations for manned missions to Mars will be 2-3 years. Astronauts will be solely responsible for implementing maintenance required on the vehicle during that time. A critical component of ensuring a successful mission is to implement vehicle design requirements that incorporate the following logistical and maintenance considerations: easy accessibility to Orbital Replacement Units (ORUs), standardized crew interfaces, standardized tools and support/diagnostic equipment, and minimal repair operations requiring Extra-Vehicular Activity (EVA). However, even with these design requirements met, operational maintenance requirements for a Mars mission will be more challenging than previous US manned spaceflight programs. Historically, US manned missions have involved relatively short duration flights, thus the vehicles mostly relied on system redundancy, provided minimum volume for tools/spares, and had limited on-orbit repair capability. This is even true of the current Space Shuttle program. However, the International Space Station (ISS), which has been human-inhabited continually since October 2000, has been the best example of implementing on-orbit maintenance on a vehicle which cannot be serviced via ground technicians. Lessons learned on the ISS program have shown that substantial amounts of crew time are required for on-orbit execution of maintenance tasks.

Since the Space Shuttle Columbia accident in 2003, it has become clear that Intermediate-level (I-level) maintenance capabilities, meaning ORU sub-component repair, are crucial for spacecraft without re-supply vehicles. However, I-level maintenance tasks tend to be much more complex and time intensive. The ISS has been the first program

demonstrating that these complex tasks can be completed. However, the lessons learned prove that there is a need for improvement in future programs. Future manned spaceflight programs, especially long duration missions such as to Mars, must consider operational tools to relieve complexity and minimize the crew time required for vehicle maintenance once on-orbit. The following operational tools should be considered for supporting maintenance:

- **Multi-media Maintenance Systems Database giving crew access to ORU source data**
- **Animated demonstration of maintenance tasks (planned and contingency) using 3-dimensional modeling.**
- **High-fidelity ground mock-up of vehicle for performing engineering evaluations and procedure validation of unplanned maintenance tasks.**
- **Extensive imagery of vehicle prior to closeout for flight.**

Developing efficient and user-friendly operational tools such as these will significantly improve the crew maintenance operations on future vehicles.

2 Introduction

In the 1950s and 60s the space race between the United States and Russia provided the impetus for extremely accelerated human spaceflight programs, which for NASA resulted in Mercury, Gemini and Apollo. Within a decade of Russia launching the first human into space, NASA astronauts were the first humans to walk on the moon. Since the Apollo program's conclusion, low-earth orbit (LEO) has been the extent of human space exploration with the US-led Space Shuttle and International Space Station (ISS) programs. While the distances traveled from Earth have not extended as far in the last several decades as in the days of the space race, 114 space shuttle missions and over 5 years of constant ISS operations have significantly increased humans' experience in space. It is imperative that the knowledge gained and lessons learned in these previous and current space programs are retained and applied to future programs.

The future of manned spaceflight is expected to return humans to the moon and eventually beyond to Mars. Design studies have shown most likely durations for manned missions to Mars will be 2-3 years¹. This will present unprecedented challenges for spaceflight operations, mostly due to the significantly greater distances and durations required for such missions. One of these challenges will be crew maintenance of the vehicle. Earlier US programs, such as Apollo, went similar long distances but for short durations. These vehicles mostly relied on system redundancy and provided little volume for tools and spare parts. Space Shuttle missions are also relatively short in duration, spending only 10-18 days on-orbit. It too relies on system redundancy as well as ground technicians for repairs once the orbiter returns to ground. On-orbit maintenance capabilities exist, but are considered minimal in the scope of this assessment. The ISS, however, is an excellent example for examining maintenance practices and applying lessons-learned to future long-duration missions. Although ISS crews are typically on-orbit no longer than 6 months at a time, the vehicle itself has components that were initially launched in 1998. The first permanent crew, Expedition 1, inhabited the ISS on November 2, 2000, and ISS crews have been maintaining the vehicle on-orbit ever since. The sparing philosophy for ISS has been primarily based on repair at the ORU level. However, certain capabilities have been developed for Intermediate-level (I-level), meaning sub-ORU component replacements. As more modules have been added to the ISS via Russian and American assembly flights, the vehicle has become more complex and therefore more maintenance has been required. Substantial maintenance experience has been gained with the ISS, including numerous complex repairs. And since the Space Shuttle Columbia accident in February 2003, ISS has experienced a major reduction in re-supply capabilities.

Obviously, it will never be the primary goal of any modern space program to simply maintain a vehicle in space. There are important exploration and scientific objectives that will be at the core of every future manned space mission we embark upon. Therefore, it is critical to minimize and optimize maintenance operations so that crew time can be utilized to support primary mission objectives. Looking at lessons learned from the ISS program, and applying them to future vehicle design and operational tools will ensure that our past experiences will be worthwhile and used to benefit future endeavors. The scope of this report will discuss Intra-vehicular Activity (IVA) maintenance of US segment only. Extra-vehicular Activity (EVA) maintenance requiring spacewalks and Russian system maintenance will not be considered.

3 ISS Maintenance Lessons Learned

In the first 5 years of ISS operations, crews have spent over 4,000 hours performing preventive and corrective IVA maintenance tasks². That averages 1.9 hours or more per work day and 1.8 hours or more per rest day performing ISS maintenance tasks, which exceeds that estimated by vehicle design⁴. A major shortcoming of the ISS program has been crew time allotted for science. Originally, the scientific research (also called “payload operations”) done aboard ISS was assumed to occur after ISS assembly and once the ISS was manned with 6-7 crewmembers. Due to substantial delays in both the ISS and shuttle programs, it was decided that payload operations would begin immediately and in parallel with assembly and maintenance, despite having only 2-3 crewmembers on-orbit.

Table I shows the number of crew hours spent during each 6-month Expedition Crew mission from the launch of the first ISS module in 1998 through the end of Expedition 11 in October 2005. Times are also divided into categories by United States Operational Segment (USOS) versus Russian Operational Segment (RSOS), and Preventive Maintenance (PM) versus Corrective Maintenance (CM). A significant increase is seen in the first few Expeditions due to ISS assembly progression. Note that during Expedition 6 the Space Shuttle Columbia accident occurred. This resulted in a halt of ISS assembly, reduced re-supply, and subsequent crews consisting of only 2 people. So although the total maintenance hours performed by Expedition 7 and beyond are in family with previous increments having the same vehicle configuration, the affect on overall crew time is much more severe since tasks could only be performed by 2 instead of 3 crewmembers.

Table 1. – Summary of ISS Crew Maintenance Time Spent On-Orbit²

Expedition	ISS Crew Size	USOS-PM	USOS-CM	RSOS-PM	RSOS-CM	Totals
	(# of people)	(hours)	(hours)	(hours)	(hours)	(hours)
0	0	2	8	1	40	51
1	3	8	3	19	14	44
2	3	24	148	138	81	391
3	3	39	19	130	13	201
4	3	63	46	206	123	438
5	3	60	102	196	45	403
6	3	55	103	211	93	462
7	2	80	28	244	53	405
8	2	64	104	184	97	449
9	2	147	101	186	58	492
10	2	73	96	186	97	452
11	2	42	46	117	70	275
Totals		657	804	1818	784	4063

This section details some of the lessons learned in performing ISS IVA maintenance tasks that are applicable when considering future long duration missions.

3.1 MTBF Inaccuracies

The Mean Time Between Failures (MTBF) is a calculation that estimates the average length of time an ORU operates without failing, and is based on ground testing³. The MTBFs for ISS ORUs are used in determining how many spares are pre-positioned on-orbit to accommodate failures. This, combined with criticality of the ORU failure, has also led to designating certain ORU spares as Launch On Need (LON), meaning the spare would be launched on the next re-supply mission after failure is detected. In general, this concept has been successful for ISS. However, for a few ORUs the MTBFs have proven to be extremely inaccurate. For example, the Node 1 Multiplexer/De-Multiplexers (MDMs) became operational on-orbit in November 1998. The estimated MTBF for those ORUs is 18,648 hours. Yet to date the MDMs have not had any failures requiring repair. In this case the MTBF has been exceeded by more than 300%. Alternatively, the cabin lighting in the USOS has experienced failures significantly more frequently than the estimated MTBF. These General Luminaire Assemblies (GLAs) have two sub-ORUs: the Lamp Housing Assembly (LHA), which houses the fluorescent light bulb, and the Base Ballast

Assembly (BBA), which houses the electronics and dimming and on/off switches. The LHAs have an estimated MTBF of 27,910 hours, yet the actual average has been 16,235 hours, and on-orbit operational data has shown a 35% chance of failure with less than 6,000 hours of operations. The BBA MTBFs were so high that they were assumed to have no failures during the operational lifetime of ISS. However, to date there have been 33 GLA failures, four of which were BBA failures⁵. It is not clear why these ORUs are failing at a faster rate on-orbit than predicted, but the overestimated MTBF has led to insufficient supply of spares on-orbit. And because re-supply capabilities to ISS have been reduced since the Space Shuttle Columbia accident, reduced lighting has impacted general crew operations at times.

MTBF calculations can be extremely useful in determining sparing needs when they are accurate. For long-duration missions without re-supply options, this becomes even more crucial. A 3-year mission to Mars needs to have adequate spares on-orbit to maintain the vehicle, yet minimize the volume used for spares. This balance can be optimized by increasing reliability of hardware, and performing extensive testing to prove performance stability when calculating MTBF. Since MTBFs have occasionally proven to be inaccurate, future programs should account for this possibility since it could lead to significant operational problems (i.e., inadequate lighting for crew operations) and increased amount of crew time spent performing corrective maintenance. This must be taken into account when developing sparing concepts for future programs.

3.2 Non-ORUs Failures Occur, Require More Complex Repairs

The ISS program designates certain equipment as ORUs, which usually defines an assembled unit which can be isolated from the rest of its sub-system and removed and replaced by the crew on-orbit. ORUs have specific design requirements which demand adequate accessibility, labeling, and tool interfaces to allow for easy crew repair on-orbit. However, there are certain pieces of ISS equipment whose MTBFs exceed the operational lifetime of ISS (15 years), thus were not designated ORUs and do not meet the design criteria of an ORU. One example of this is the installation fasteners on MDMs. Several years after ISS was operational, it was determined that these fasteners had a limited-life cycle and would have to be replaced after a certain number of ORU change-outs. To prevent having to return an otherwise good MDM chassis to the ground, an on-orbit procedure was developed to replace the old fasteners. However, crew evaluation of the procedure determined the task was an extremely difficult because the hardware was not designed to be repaired on-orbit, and it required handling multiple, extremely small parts in micro-gravity.

Another example is the Suit Processing Cooling Unit (SPCU), which is installed in the USOS Airlock and used as a heat exchanger that interfaces with the EVA suits the crew wears during spacewalks. The SPCU had a failure on-orbit, which prohibited EVAs out of the USOS Airlock. The SPCU was originally designated as an ORU. However, the design did not adhere to the associated requirements because on-orbit maintenance was not expected to occur. As a result, the maintenance procedure was extremely complex as it involved cutting away insulation foam that had been permanently installed with Room Temperature Vulcanization (RTV), which generated a significant amount of debris. In addition, the crew had to handle non-captive fasteners, nuts and washers and removal of permanent clamps. In total, the procedure took approximately 17 hours of on-orbit crew time to complete⁶.

ISS operations have demonstrated that equipment not designated as ORUs do occasionally require on-orbit repairs. Experience indicates that this often results in a complex, time-intensive on-orbit crew repair. For future programs, it is recommended that any equipment with failure mechanisms be under the same design requirements as an ORU to minimize repair complexities as much as possible.

3.3 Unexpected Failures/Workarounds Require Expansive Complement of Tools/Materials

The ISS has demonstrated the importance of providing an expansive complement of tools, materials and equipment to respond to unexpected failures and hardware workarounds. The ISS repair tools include IVA hand tools, a Pin Kit for wire splicing and connector repair, Seal Kit for replacing fluid line hard and soft seals, Leak Pinpoint and Repair Kit, and a Clamp and Bracket Kit. The follow examples show how these some of these tools and materials were used to repair unexpected hardware anomalies on-orbit:

- A very small cabin leak was detected in the USOS Lab module. An Ultrasonic Leak Detector was used to isolate the leak to a flex hose that fed the volume between Lab window panes to vacuum. The flex hose had been inadvertently damaged because it was protruding into the crew translation area. The flex hose was eventually replaced, but to prevent any future damage, a protective metal box was constructed using on-orbit materials in the Clamp & Bracket Kit.

- When installing Active Rack Isolation System (ARIS) hardware (a modification kit to a payload rack), the crew found a metallic part was not able to be installed due to tolerance issues. A file was used to reduce the diameter of the part, and it was subsequently installed without needing to launch a replacement.
- The Carbon Dioxide Removal Assembly in the US Laboratory module was experiencing unexpected failures, thus an upgrade to install sock filters on the lines to prevent Zeolite contamination was performed. This required manually accessing valves with fittings torqued to extremely high values. Despite extensive procedure validation using ground equipment, the crew was unable to loosen one of the valves on-orbit. Mission Control Center in Houston suggested re-attempting with a different tool, a strap wrench, which had never been required for use before. This tool easily solved the problem, and the maintenance procedure was completed successfully.

ISS operations have proven that having a vast complement of tools and repair kits is required for responding to unexpected hardware failures and maintaining vehicle integrity. This philosophy should be implemented in future programs, especially since re-supply options will be minimal or non-existent.

3.4 Overly Complex Diagnostic Tools Hinder Simple Tasks

Diagnostic tools are useful in diagnosing hardware failures on-orbit. The most commonly used of these tools on ISS, the Fluke Scopemeter, can be used to measure temperature, pressure and current. This tool can perform extremely complex measurements and has recording/downloading features. While these capabilities are useful to have, they are not typically required while using the Scopemeter for standard tasks. Many ISS crews have raised concerns regarding the complexity of using this tool for performing simple procedures such as pressure checks. Thus, a simpler multi-meter was scavenged from the Space Shuttle during the joint STS-114 mission so that ISS crews could simplify these tasks.

Although “all-in-one”-type diagnostic tools minimize stowage volume required, it is recommended that tools with varying complexity are included in future long-duration missions. This will allow simple tasks to be performed more efficiently, without significant overhead time required to operate tools designed for more complex tasks.

3.5 Lack of Re-supply Affects Consumables, Limited-Life Items and Calibrated Equipment

The ISS includes numerous materials and equipment on-orbit that require re-supply. With the Space Shuttle Columbia accident in 2003, re-supply resources to ISS were significantly reduced. Ever since, re-supply has relied solely upon Russian Progress and Soyuz vehicles, which have significantly less volume available compared to the Orbiter middeck and the Multi-Purpose Logistics Module (MPLM) available on Shuttle logistics flights. Thus, there have been significant effects on the following maintenance-related categories of equipment which rely on re-supply:

- *Consumables* used for maintenance tasks such as duct tape and non-rechargeable batteries had to be rationed until re-supply occurred.
- *Limited-life Items* such as crew cabin air filters and seals were required to operate beyond expected life cycle. Duxseal, a putty-like substance to be used to patch a hole in the module wall to stop cabin pressure leakage, is another limited-life item, but since it supports crew and vehicle safety, it was considered high-priority for re-supply missions. Although no specific problems were encountered, operating equipment beyond the life-limit not a preferred practice from an engineering perspective as it could result in reduced equipment performance and increased risk of system failure.
- *Calibrated Equipment* such as torque wrenches and pressure probes were required to operate beyond their calibration certification dates. Additional crew maintenance time was required to perform accuracy validation procedures of the associated tools before each use.

Equipment with limited-life and calibration issues will be a challenge for long-duration missions without re-supply. Hardware engineers will need to develop ways to design and certify equipment to allow a longer operation lifetime, ideally beyond the mission duration. For items with life-cycle issues, design robustness must be increased, or redundant hardware will have to be provided, which impacts stowage volume. On-orbit calibration techniques will also need to be developed.

3.6 I-Level Maintenance can Reduce Required Stowage Volume

Stowage volume is a major constraint for ISS. There is far more loose equipment on-board than designated stowage volume to hold it. Thus the crew has had to utilize alternative volumes for stowing equipment, which has impacted operations. For example, when preparing for a Space Shuttle logistical mission, pre-packed equipment is stowed in front of a closed hatch where the upcoming MPLM will be installed. However, any maintenance tasks

that require access to that area require crew to move all the stowed equipment out of the way, and then replace it after the task. This practice wastes a considerable amount of crew time. ISS spares take up a significant portion of stowage volume. One way to minimize this would be to implement a sub-ORU level sparing concept. This is known as Intermediate-level, or I-level maintenance. ISS has utilized I-level maintenance for the USOS computers, MDMs. The MDMs are comprised of a chassis, and circuit cards. Although there are many types of MDMs on-orbit, they use common components, so fewer large chassis spares are required by manifesting these circuit card sub-components. Applying this I-level repair approach to other equipment, can increase efficiency by replacing only the failed components instead of entire ORUs, and thus maximize stowage volume for spares.

3.7 Commonality Increases Operational Efficiency (crew interfaces, fastener size, batteries, etc.)

ISS is comprised of several different pressurized modules designed and manufactured by multiple international partners. Although interface control documents were implemented to ensure that the different modules could interface at a sub-system level, there are many differences in crew interfaces between the modules. For example, almost all USOS equipment uses fasteners sized in English units, whereas the RSOS uses metric. The European Laboratory utilizes unique fasteners with a star-shaped interface, so unique tools are required. The result is that ISS contains 3 separate tool kits to accommodate all equipment on-board. For ISS this is a bi-product of an international project. But, it still demonstrates that commonality should be applied wherever practical to reduce the number of tools required. In addition, using common chassis and fastener design among ORUs where practical will also optimize maintenance operations, as this minimizes tools required and increases crew familiarity.

Another aspect of commonality is rechargeable batteries and power supply equipment. ISS utilizes many different types of batteries for operating equipment such as power tools, digital cameras, flashlights, etc. In addition, the RS and USOS utilize different power voltages (28V versus 120V, respectively), so unique power supplies, converters and cables exist for each. Requiring battery-powered or plug-in type equipment to conform to a common set of batteries and power equipment for future long-duration missions will minimize the amount crew time spent on battery charging and equipment setup activities. This can also be beneficial from a stowage perspective, as discussed in Section 3.6.

It is recognized that hardware commonality can also lead to design inefficiencies. For example, if all electronics boxes are required to use the same fasteners, this may not result in optimal geometric design and mass-savings. And of course, not all powered equipment will be capable of running on the same batteries. However, future long-duration missions in the future should consider equipment commonality as much as practical in the design phase to improve operational efficiency.

4 Operational Tools for Supporting Crew Maintenance Tasks

In future manned spaceflight programs, especially long duration missions such as to Mars, it will be impossible for pre-flight training to cover every possible maintenance scenario the crew may encounter. Therefore, a skills-based approach should be used instead of task-based. In addition, one must consider operational tools to relieve complexity and minimize the crew time required for vehicle maintenance once on-orbit. The following operational tools should be considered for supporting maintenance in future programs:

4.1 Multi-media Maintenance Systems Database

For future long-duration missions, it will be more imperative than ever that crews be intimately familiar with the vehicle they are operating. From a maintenance perspective, this means knowing the vehicle hardware configuration and being able to utilize various data sources and implement successful maintenance tasks. Due to delayed communication over long distances, it may not be always be possible to uplink data to the crew. Therefore, it would be optimal to have all ORU source data available on-board for reference. This would include data such as illustrated parts break-downs, engineering drawings, ORU maintenance task source data, etc. These references would be easiest to decipher by providing a common database for all maintenance data, and linking them from the operational procedures the crew would use for performing the tasks. This database could also contain user-changeable information such as maintenance logs for tracking ORU maintenance histories and stowage information for tools, spares, etc. Having all this information in a single database would allow crew to be more self-sufficient (i.e., not so dependent on the ground for data), and more efficient during maintenance operations.

4.2 Animated Demonstration of Maintenance Tasks for On-board Training

Providing short animations demonstrating maintenance tasks would be extremely useful for the crew to review a previously trained task, or preview a complex maintenance task prior to performing the activity on-orbit. Considering that 3D-CAD models are already being used for spacecraft design, the development of this tool would be relatively simple and could be incorporated into a maintenance database as described in Section 4.1 for all planned and contingency maintenance tasks a crew could encounter. Having this capability would increase efficiency, performance and success for crew maintenance tasks.

4.3 High-fidelity ground mock-up of vehicle

It is imperative that a high-fidelity mock-up of future manned spacecraft be available for performing engineering evaluations and procedure validation of unplanned maintenance tasks. It is inevitable that unexpected anomalies will occur during the mission, and ground engineers will need an accurate mock-up for developing and validating contingency maintenance tasks prior to crew execution.

4.4 Extensive imagery of vehicle prior to closeout for flight

Another important operational tool during a mission is pre-flight imagery of the vehicle. While a vehicle mock-up can be extremely useful for emulating the flight vehicle, certain details such as specific wire cable routing, or locations of labels, have been difficult to precisely duplicate. Therefore, extensive imagery of the vehicle should be collected prior to launch. This would be an invaluable tool for ground engineers, and could also be a useful resource for crew viewing of hardware configurations behind closeouts or for comparison should there be any physical changes to the vehicle throughout the mission. It is highly recommended that the operations community be involved in the pre-flight imagery collection to ensure that the imagery taken is useful in an operational scenario.

Developing efficient and user-friendly operational tools such as those mentioned above will significantly improve crew maintenance operations of future vehicles.

5 Conclusion

Table II summarizes lessons learned from ISS crew maintenance operations, and the associated recommendations for future long-duration missions.

Table II. ISS Lesson Learned and Recommendations

ISS Lesson Learned	Recommendation for future Long Duration Missions
MTBF Inaccuracies	Increase MTBF accuracy by improving hardware reliability and testing for proof of performance. Sparing concept should not assume MTBFs entirely accurate, and allow for margin since re-supply assumed not available.
Non-ORU Failures Occur, Require More Complex Repairs	Apply equal design requirements for equipment not expected to fail as those expected to fail as much as practical to reduce complexity of unexpected repairs.
Unexpected Failures Require Expansive Complement of Tools/Materials	Implement philosophy similar to ISS to manifest a large variety of diagnostic and repair tools to allow for optimal operational capabilities.
Overly Complex Diagnostic Tools Hinder Simple Tasks	Manifest diagnostic tools with varying complexity to meet operational needs.
Lack of Re-supply Affects Consumables, Limited-Life Items, and Calibrated Equipment	Design/certify equipment with to maximize life-limits beyond mission duration. Increase hardware robustness to maximize life-cycle limits or provide redundant hardware. Develop on-orbit calibration techniques.
I-level Maintenance Capabilities can Reduce Required Stowage Volume	Design ORUs with I-level maintainability so that only failed components are replaced to increase sparing efficiency and reduce required stowage volume.
Commonality Increases Operational Efficiency	Design common fasteners to reduce tool kit requirements. Maximize commonality for rechargeable batteries and plug-in equipment.

The following operational tools are recommended to optimize efficiency of on-orbit maintenance:

- Multi-media Maintenance Database
- Animated Demonstration of Maintenance Tasks for On-board Training
- High-fidelity Ground Mock-Up of Vehicle
- Extensive imagery of vehicle prior to closeout for flight

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7 References

1. Zubrin, Robert. The Case For Mars: The Plan to Settle the Red Planet and Why We Must, Free Press, New York, 1997.
2. ISS Maintenance Summary, unofficial NASA document.
3. International Space Station Familiarization (TD9702A), July 31, 1998.
4. "Applying Analysis of International Space Station Crew-Time Utilization to Mission Design", James. F Russell, David M. Klaus, Todd J. Mosher. Journal of Spacecraft and Rockets, Vol. 43. No. 1, January-February 2006.
5. Designated Maintenance Items (DMI) List, maintained by NASA-JSC/DF53, March 2006 version.
6. Station Operational Data File (SODF), SPCU R&R.